

PartnerAPI

Audited Documentation Addendum

Software Engineering Project

October 6, 2026

Contents

1	Introduction	2
2	Testing Addendum (Supplement to Test Plan)	2
2.1	Defect Log	2
2.2	Defect Density Computation	2
2.3	Defect Removal Efficiency (DRE)	2
3	Project Plan Addendum	4
3.1	Network Diagram (PERT)	4
3.2	Gantt Chart	4
4	Risk Register Addendum	5
4.1	Probability-Impact Matrix	5
5	Summary	5

1 Introduction

This document serves as the formal addendum to the PartnerAPI project documentation suite. It supplements the original documents by providing the explicitly required visual diagrams and computed mathematical evidence for testing and risk management as highlighted during the documentation audit.

2 Testing Addendum (Supplement to Test Plan)

2.1 Defect Log

Table 1 provides the aggregated defect counts identified during the initial testing phases of PartnerAPI.

Metric	Count
Defects found before release (D_{before})	25
Defects found after release (D_{after})	2
Total Defects Identified (D_{total})	27

Table 1: PartnerAPI Summary Defect Log

2.2 Defect Density Computation

Defect Density measures the number of defects relative to the size of the software.

For PartnerAPI, the software size is evaluated at 10 KLOC (Thousands of Lines of Code).

$$\text{Defect Density} = \frac{\text{Total Defects}}{\text{Software Size (KLOC)}}$$

Therefore:

$$\text{Defect Density} = \frac{27}{10} = \mathbf{2.7} \text{ defects per KLOC}$$

2.3 Defect Removal Efficiency (DRE)

Defect Removal Efficiency determines the percentage of defects successfully caught prior to production release.

$$\text{DRE} = \left(\frac{D_{\text{before}}}{D_{\text{before}} + D_{\text{after}}} \right) \times 100$$

Therefore:

$$\text{DRE} = \left(\frac{25}{25 + 2} \right) \times 100$$

$$\text{DRE} = \frac{25}{27} \times 100 = \mathbf{92.59\%}$$

A DRE of 92.59% indicates that the majority of identified defects were detected before production release.

3 Project Plan Addendum

3.1 Network Diagram (PERT)

The network diagram below illustrates the sequential critical path of the standard baseline release. Custom partner requests are excluded from this baseline path.



Figure 1: PartnerAPI Network Diagram (Critical Path)

3.2 Gantt Chart

The Gantt chart visually maps the schedule timeline against the three-point Expected Time (TE) values calculated in the Estimation Sheet.

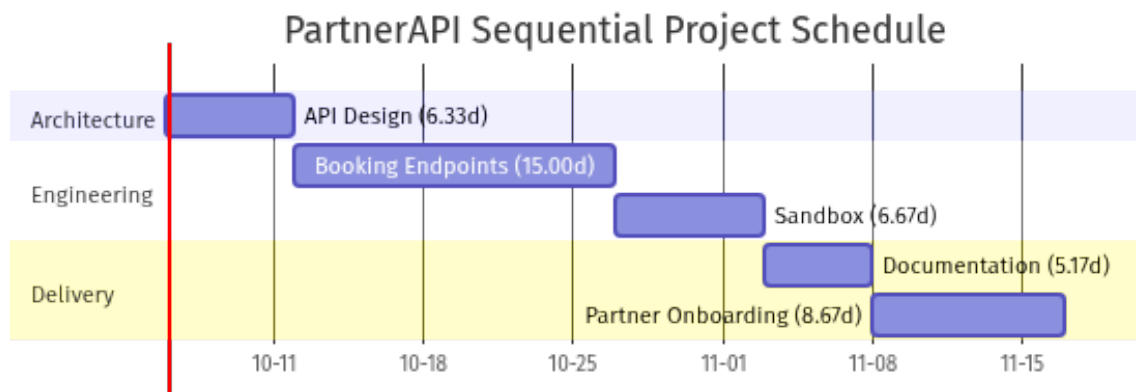


Figure 2: PartnerAPI Baseline Gantt Chart

4 Risk Register Addendum

4.1 Probability-Impact Matrix

Table 2 visualizes the standard Probability-Impact Matrix utilized to assess Risk Exposure for PartnerAPI.

The risks R1 through R5 from the Risk Register are plotted according to their assessed severity.

		Impact		
		Low	Medium	High
Probability	High			R1, R3
	Medium		R5	R2
	Low			R4

Table 2: PartnerAPI Probability-Impact Matrix

Legend:

- **R1**: Internal booking-system integration delay (High/High).
- **R2**: Partner integration problems (Medium/High).
- **R3**: Scope expansion through custom requests (High/High).
- **R4**: Security vulnerability (Low/Very High, mapped to High for the standard grid).
- **R5**: Documentation delay (Medium/Medium).

5 Summary

The audited addendum provides additional quantitative and visual evidence supporting the PartnerAPI project documentation.

The testing evidence records 27 identified defects, consisting of 25 defects found before release and 2 defects found after release. This results in a defect density of 2.7 defects per KLOC and a Defect Removal Efficiency of 92.59%.

The network diagram and Gantt chart provide visual evidence of the baseline project schedule, while the Probability-Impact Matrix summarises the assessed project risks.

These additions complement the existing Test Plan, Project Plan and Risk Register documentation.